High Level Design Document

Introduction

This High Level Design (HLD) document outlines the architecture and core components for **LoanForest - Loan Default Predictor**. The project delivers a Streamlit web application that predicts loan default risk using a Random Forest machine learning model, allowing users to interactively input features via sliders.

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1. System Architecture Overview

Architecture Description:
The system is a single-page web application built with Streamlit, integrating a pre-trained Random Forest model (scikit-learn) for real-time loan default prediction. The app runs on a Python backend, handling user input, model inference, and result display.

Module/Component	Role/Responsibility
User Interface (UI)	Collects user input via sliders, displays predictions
Model Inference Layer	Loads and applies Random Forest model to input features
Data Preprocessing	Formats and validates user input for the model
Streamlit App Server	Hosts the app, manages UI and backend integration

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2. Component Interactions

Sequence Step	Interaction Description
1	User adjusts feature sliders in the Streamlit UI
2	UI sends input data to the backend (Streamlit server)
3	Data Preprocessing module validates and formats input
4	Model Inference Layer applies Random Forest model to processed input
5	Prediction result is returned to the UI and displayed to the user

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3. Data Flow Overview

Source	Data	Destination	Purpose
User	Feature values	UI	Collect loan applicant details
UI	User input	Data Preprocessing	Prepare data for model
Preprocessing	Formatted features	Model Inference Layer	Predict default risk

Model	Prediction result	UI	Display risk to user

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4. Technology Stack

Layer/Function	Technology/Framework
Web UI & Server	Streamlit (Python)
Machine Learning	scikit-learn (RandomForest)
Data Processing	Python (pandas, numpy)
Deployment	Streamlit Cloud / Local

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5. Scalability & Reliability

- **Scalability:** Suitable for single-user or small team use; can be containerized or deployed on Streamlit Cloud for broader access.
- **Reliability:** Stateless design; model and app can be reloaded on restart. Input validation ensures robust predictions.
- **Security:** No sensitive data stored; all processing is in-memory per session.

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